시계열분석

6주차: ARIMA 모형의 식별 및 예측

AR(2) Process

- AR(2): $Z_t = \phi_1 \cdot Z_{t-1} + \phi_2 \cdot Z_{t-2} + \varepsilon_t$

- $(1 - \phi_1 B - \phi_2 B^2)\rho_k = 0$

$\lambda^2 - \phi_1 \lambda - \phi_2 = 0$, $\lambda = \begin{cases} \lambda_1, \lambda_2 & \text{서로 다른 실근} \\ \lambda_1 = \lambda_2 = \lambda & \text{중근} \\ \lambda_1, \lambda_2 & \text{복소수근} \end{cases}$

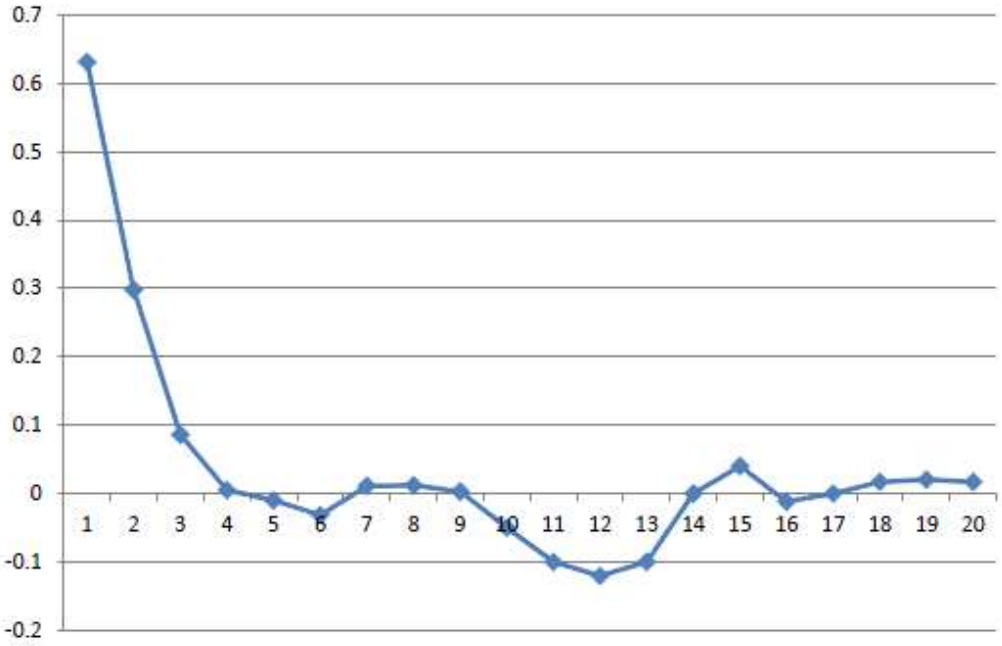
AR(2) Process

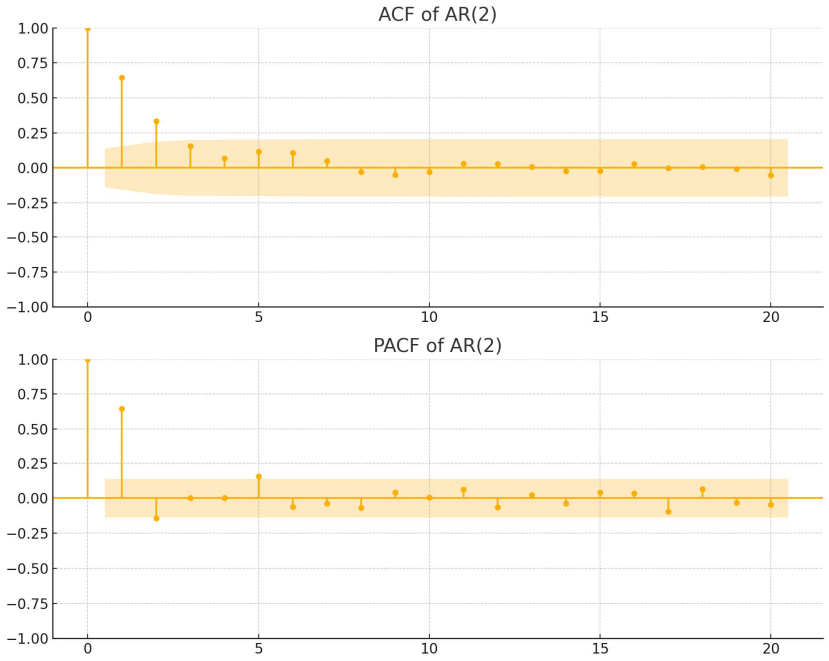
1. 두 개의 실근

$$Z_t = 0.8Z_{t-1} - 0.15Z_{t-2} + \varepsilon_t$$

$$\rho_k = c_1\lambda_1^k + c_2\lambda_2^k$$

$$\lambda_1 = 0.5,\ \lambda_2 = 0.3$$





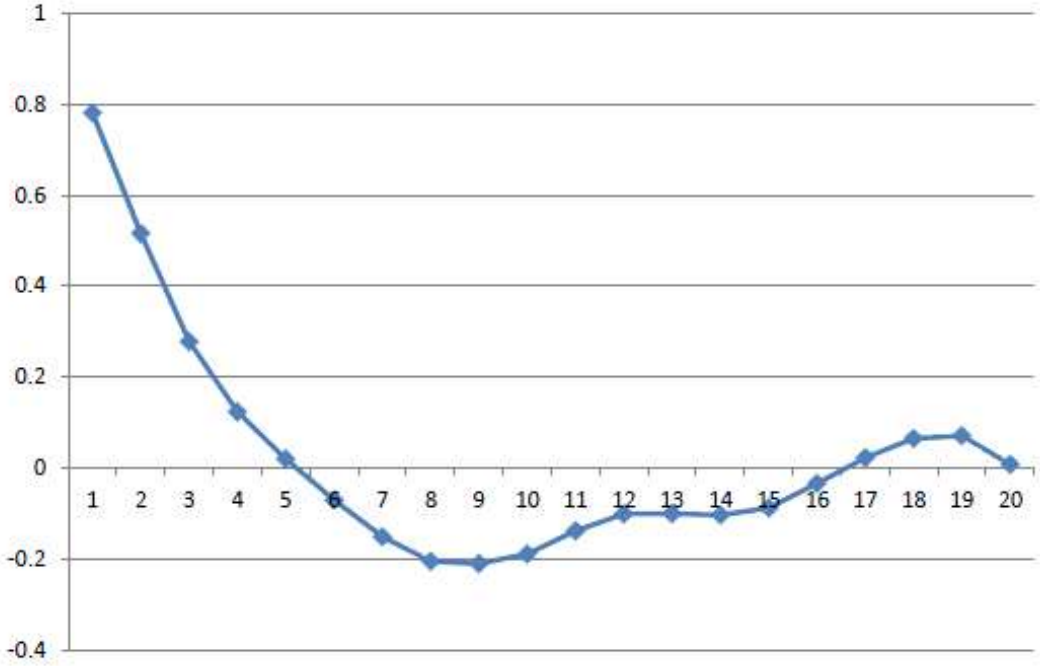
AR(2) Process

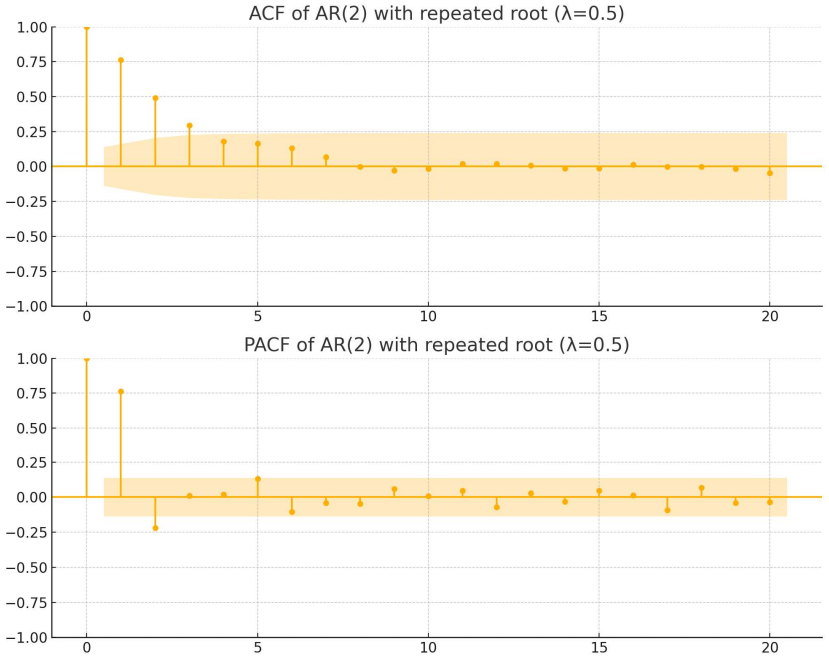
$$Z_t = Z_{t-1} - 0.25Z_{t-2} + \varepsilon_t$$

$$\rho_k = c_1\lambda_1^k + c_2(k\lambda_1^k)$$

$$\lambda_1 = 0.5$$

2. 한 개의 실근(중근)





AR(2) Process

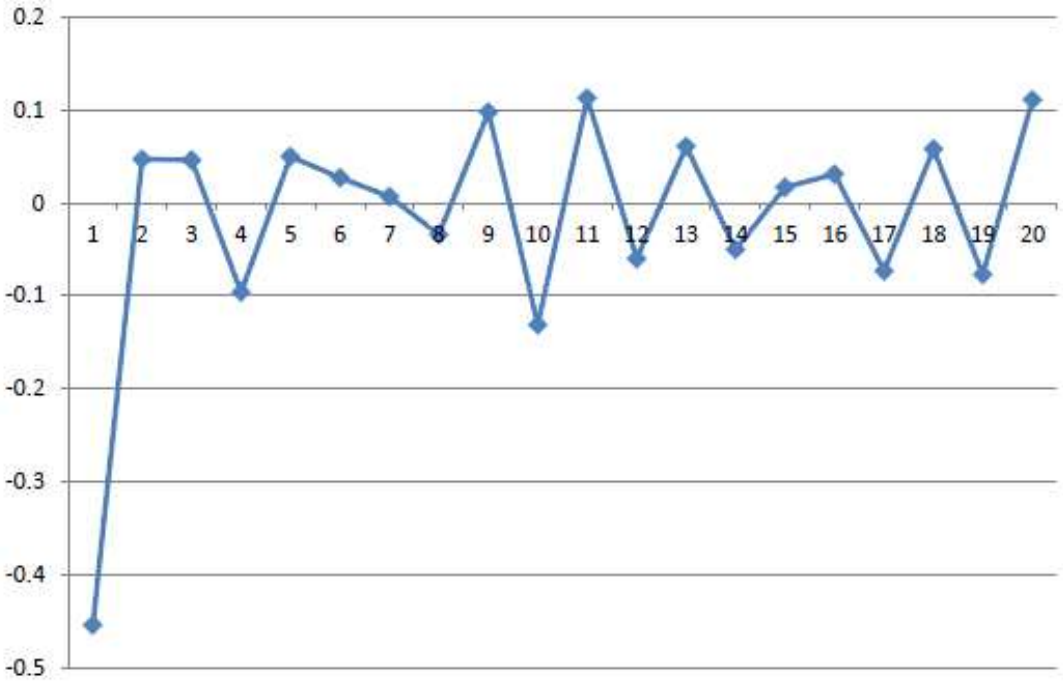
3. 두 개의 허근

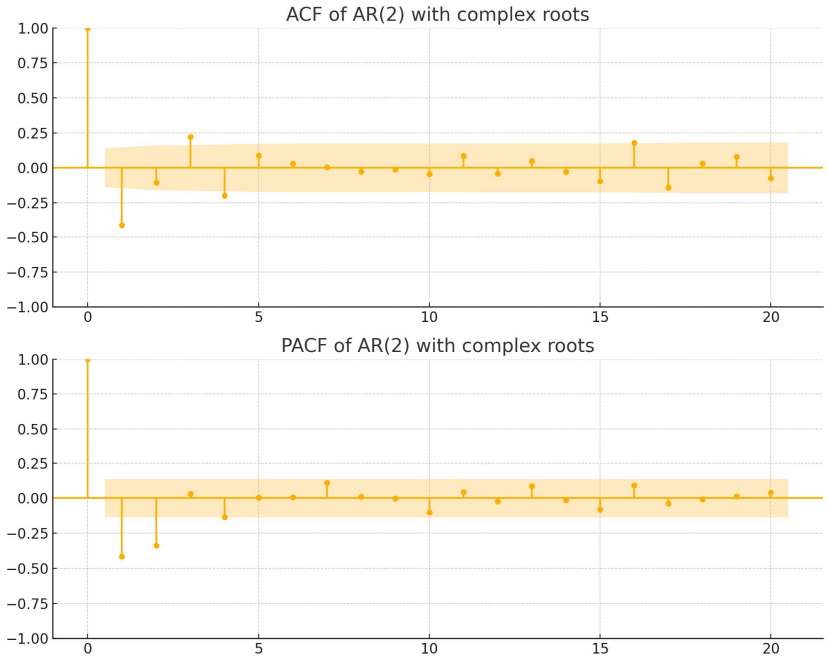
$$Z_t = -0.5Z_{t-1} - 0.3Z_{t-2} + \varepsilon_t$$

$$\rho_k = c_1\lambda_1^k + c_2\lambda_2^k$$

$$\lambda_1 = \frac{-0.5 + \sqrt{0.95}i}{2}$$

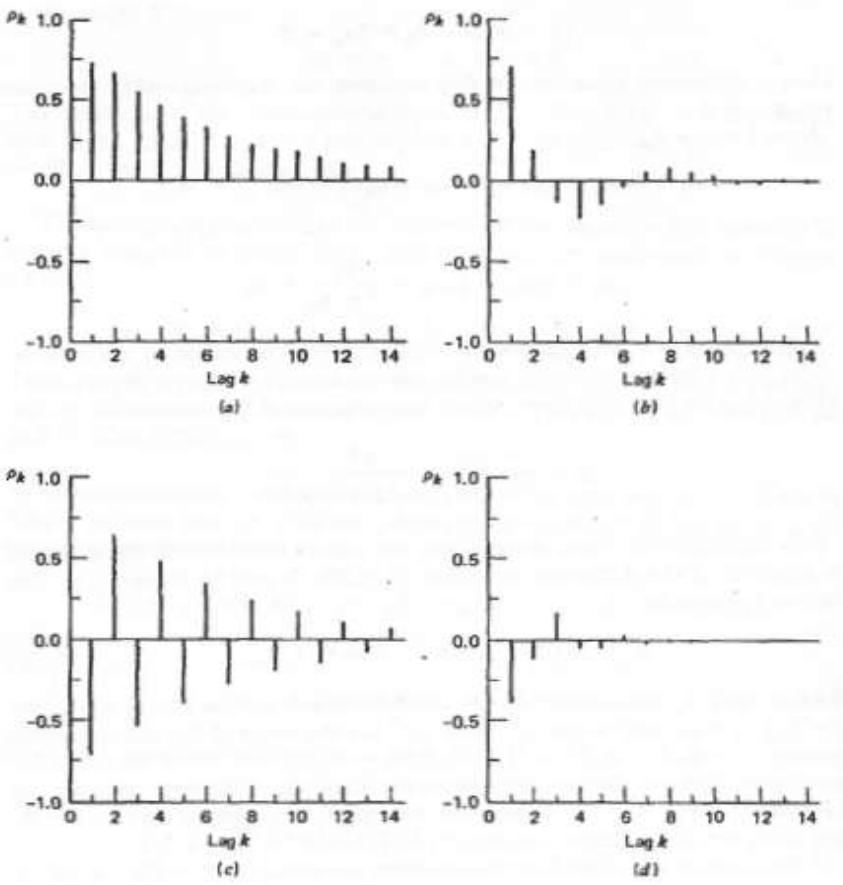
$$\lambda_2 = \frac{-0.5 - \sqrt{0.95}i}{2}$$





AR(2) Process

그래프	수식 예시	근의 형태	ACF 형태 요약
(a)	$Z_t = 0.5Z_{t-1} + 0.3Z_{t-2} + \varepsilon_t$	두 실근	지수감쇠, 양수 유지
(b)	$Z_t = Z_{t-1} - 0.5Z_{t-2} + \varepsilon_t$	복소근 (양의 실수부)	감쇠 진동, 느린 감소
(c)	$Z_t = -0.5Z_{t-1} + 0.3Z_{t-2} + \varepsilon_t$	두 실근 (음수)	진동, 감쇠
(d)	$Z_t = -0.5Z_{t-1} - 0.3Z_{t-2} + \varepsilon_t$	복소근 (음의 실수부)	진동 + 빠른 감쇠



$$(a)\ z_t = 0.5z_{t-1} + 0.3z_{t-2} + \varepsilon_t$$

$$\rightarrow \lambda_1 = \frac{0.5+\sqrt{1.45}}{2}, \lambda_2 = \frac{0.5-\sqrt{1.45}}{2}$$

$$\Rightarrow \rho_k = c_1\left(\frac{0.5+\sqrt{1.45}}{2}\right)^k + c_2\left(\frac{0.5-\sqrt{1.45}}{2}\right)^k$$

$$(b)\ z_t = z_{t-1} - 0.5z_{t-2} + \varepsilon_t$$

$$\rightarrow \lambda_1 = \frac{1+i}{2}, \lambda_2 = \frac{1-i}{2}$$

$$\Rightarrow \rho_k = c_1\left(\frac{1+i}{2}\right)^k + c_2\left(\frac{1-i}{2}\right)^k$$

$$(c)\ z_t = -0.5z_{t-1} + 0.3z_{t-2} + \varepsilon_t$$

$$\rightarrow \lambda_1 = \frac{-0.5+\sqrt{1.45}}{2}, \lambda_2 = \frac{-0.5-\sqrt{1.45}}{2}$$

$$\Rightarrow \rho_k = c_1\left(\frac{-0.5+\sqrt{1.45}}{2}\right)^k + c_2\left(\frac{-0.5-\sqrt{1.45}}{2}\right)^k$$

$$(d)\ z_t = -0.5z_{t-1} - 0.3z_{t-2} + \varepsilon_t$$

$$\rightarrow \lambda_1 = \frac{-0.5+\sqrt{0.95i}}{2}, \lambda_2 = \frac{-0.5-\sqrt{0.95i}}{2}$$

$$\Rightarrow \rho_k = c_1\left(\frac{-0.5+\sqrt{0.95i}}{2}\right)^k + c_2\left(\frac{-0.5-\sqrt{0.95i}}{2}\right)^k$$

Partial Autocorrelation for AR(p)

- $Z_t = \phi_{11} \cdot Z_{t-1} + \varepsilon_t$
- $Z_t = \phi_{21} \cdot Z_{t-1} + \phi_{22} \cdot Z_{t-2} + \varepsilon_t$
- $Z_t = \phi_{31} \cdot Z_{t-1} + \phi_{32} \cdot Z_{t-2} + \phi_{33} \cdot Z_{t-3} + \varepsilon_t$
- $\vdots$

Partial Autocorrelation for AR(p)

- $\phi_{22} = \text{corr}\left[Z_t - E[Z_t|Z_{t-1}], Z_{t-2} - E[Z_{t-2}|Z_{t-1}]\right]$

$$= \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}$$

Autoregressive and Moving Average Process

- ARMA(1,1)

$$Z_t = \phi \cdot Z_{t-1} + \varepsilon_t - \theta \varepsilon_{t-1}$$

- MA representation

$$Z_t = (1 - \phi B)^{-1}(1 - \theta B)\varepsilon_t$$

AR representation

- $Z_t = \phi \cdot Z_{t-1} + \varepsilon_t - \theta \varepsilon_{t-1}$
- $(1 - \theta B)^{-1}(1 - \phi B) Z_t = \varepsilon_t$
- $(1 + \theta B + \theta^2 B^2 + \theta^3 B^3 + \cdots)(1 - \phi B) Z_t = \varepsilon_t$
- $(1 + (\theta - \phi) B + \theta(\theta - \phi) B^2 + \theta^2(\theta - \phi) B^3 + \cdots) Z_t = \varepsilon_t$: $AR(\infty)$

ARMA(1,1)

- ARMA(1,1): $Z_t = \phi \cdot Z_{t-1} + \varepsilon_t - \theta \varepsilon_{t-1}$
- ACF of ARMA(1,1)

- $(1 - \phi B)\rho_k = 0, \ k \geq 2$

Forecasting Function of AR(1)

$$
\begin{aligned}
Z_{t+k} &- \phi \cdot Z_{t+k-1} &&= \varepsilon_{t+k} \\
Z_{t+k-1} &- \phi \cdot Z_{t+k-2} &&= \varepsilon_{t+k-1} \\
&\vdots \\
Z_{t+1} &- \phi \cdot Z_{t} &&= \varepsilon_{t+1}
\end{aligned}
$$

$$Z_{t+k} = \phi^{k} \cdot Z_t + \{\varepsilon_{t+k} + \phi\varepsilon_{t+k-1} + \ \ldots + \phi^{k-1}\varepsilon_{t+1}\}$$

$$f_{t,k} = E[Z_{t+k}/z_t, z_{t-1}, \ldots] =$$

$$V_{t,k} = V[Z_{t+k}/z_t, z_{t-1}, \ldots] =$$

Forecasting Function of AR(1)

$$
\begin{aligned}
(Z_{t+k}-\mu) \quad &- \phi \cdot (Z_{t+k-1} - \mu) \quad = \varepsilon_{t+k} \\
(Z_{t+k-1} - \mu) &- \phi \cdot (Z_{t+k-2} - \mu) \quad = \varepsilon_{t+k-1} \\
&\vdots \\
(Z_{t+1} - \mu) &- \phi \cdot (Z_t - \mu) \quad = \varepsilon_{t+1}
\end{aligned}
$$

$$(Z_{t+k} - \mu) = \phi^k \cdot (Z_t - \mu) + \{\varepsilon_{t+k} + \phi\varepsilon_{t+k-1} + \ \ldots + \phi^{k-1}\varepsilon_{t+1}\}$$

$$f_{t,k} = E[Z_{t+k}/z_t, z_{t-1}, \ldots] =$$

$$V_{t,k} = V[Z_{t+k}/z_t, z_{t-1}, \ldots] =$$

Forecasting Function of AR(2)

$$Z_{t+k} = \phi_1 Z_{t+k-1} + \phi_2 Z_{t+k-2} + \varepsilon_{t+k}$$

$$(1 - \phi_1 B - \phi_2 B^2) f_{t,k} = 0$$

Forecasting Function of AR(1,1)

$$Z_{t+k} = \phi Z_{t+k-1} + \varepsilon_{t+k} - \theta \varepsilon_{t+k-1}$$

$$(1 - \phi B) f_{t,k} = 0, for\ k \geq 2$$

Estimation of AR(1)

- $OLS: Z_t = \phi_0 + \phi_1 Z_{t-1} + \varepsilon_t$

$$Minimize \sum(Z_t - \emptyset_0 - \emptyset_1 Z_{t-1})^2$$

- $MLE(Maximum\ Likelihood\ Estimation)$

Random Walk

$$Z_{t+k} = Z_{t+k-1} + \varepsilon_{t+k}$$

$$Z_{t+k-1} = Z_{t+k-2} + \varepsilon_{t+k-1}$$

$$\vdots$$

$$Z_{t+1} = Z_t + \varepsilon_{t+1}$$

$$Z_{t+k} = Z_t + \{\varepsilon_{t+k} + \varepsilon_{t+k-1} + \ \ldots + \varepsilon_{t+1}\}$$

Random Walk with a Drift

$$Z_{t+k} = Z_{t+k-1} + \mu + \varepsilon_{t+k}$$

$$Z_{t+k-1} = Z_{t+k-2} + \mu + \varepsilon_{t+k-1}$$

$$\vdots$$

$$Z_{t+1} = Z_t + \mu + \varepsilon_{t+1}$$

$$Z_{t+k} = Z_t + \mu \cdot k + \{\varepsilon_{t+k} + \varepsilon_{t+k-1} + \ \ldots + \varepsilon_{t+1}\}$$

ARIMA 예제

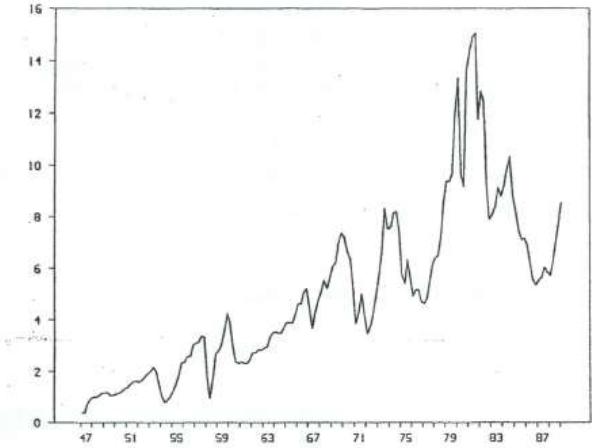


FIGURE 17.2 U.S. nominal interest rate on 3-month Treasury bills, data sampled quarterly but quoted at an annual rate, 1947:1 to 1989:1.

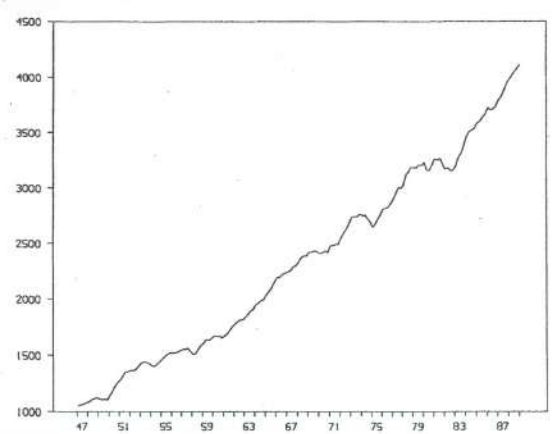


FIGURE 17.3 U.S. real GNP, data sampled quarterly but quoted at an annual rate in billions of 1982 dollars, 1947:1 to 1989:1.

$H_0: Z_t = Z_{t-1} + \varepsilon_t$ (O)

$H_1: Z_t = \alpha + \phi Z_{t-1} + \varepsilon_t$ (X)

$H_0: Z_t = \alpha + Z_{t-1} + \varepsilon_t$ (O)

$H_1: Z_t = \alpha + \delta t + \phi Z_{t-1} + \varepsilon_t$ (X)

AR(1) 시뮬레이션

```
set.seed(123)
n <- 100
phi1 <- 0.7
eps <- rnorm(n)
y_ar1 <- numeric(n)
y_ar1[1] <- 10  # 초기값
for (t in 2:n) {
  y_ar1[t] <- phi1 * y_ar1[t-1] + eps[t]
}
plot(y_ar1, type = "l", main = "Simulated AR(1) Process")
```

AR(2) 시뮬레이션

```
set.seed(123)
n <- 100
phi1 <- 0.5; phi2 <- -0.2
eps <- rnorm(n)
y_ar2 <- numeric(n)
y_ar2[1:2] <- c(10, 12)  # 초기값 설정
for (t in 3:n) {
  y_ar2[t] <- phi1 * y_ar2[t-1] + phi2 * y_ar2[t-2] + eps[t]
}
plot(y_ar2, type = "l", main = "Simulated AR(2) Process")
```

Random Walk 시뮬레이션

```
set.seed(123)
n <- 100
eps <- rnorm(n)
y_rw <- numeric(n)
y_rw[1] <- 10
for (t in 2:n) {
  y_rw[t] <- y_rw[t-1] + eps[t]
}
plot(y_rw, type = "l", main = "Simulated Random Walk")
```

Random Walk with a Drift 시뮬레이션

```
set.seed(123)
n <- 100
drift <- 0.5
eps <- rnorm(n)
y_rwd <- numeric(n)
y_rwd[1] <- 10
for (t in 2:n) {
  y_rwd[t] <- y_rwd[t-1] + drift + eps[t]
}
plot(y_rwd, type = "l", main = "Simulated Random Walk with a Drift")
```

ARIMA 모형

- Autoregressive Integrated Moving Average: 비정상 시계열 데이터를 정상화한 후 자기회귀(AR)와 이동평균(MA) 모형을 결합하여 예측하는 모형
- ARIMA(p,d,q)로 표기
  - p (자기회귀 차수, AR order): 모델이 과거 몇 개의 시점의 값을 참조하는지
  - d (차분 차수, Integration order): 몇 번의 차분을 통해 원시 시계열 y_t를 정상 시계열로 변환하는지
  - q (이동평균 차수, MA order): 모델이 과거 몇 시점의 오차를 고려하는지

$$\phi(B)(1-B)^d y_t = c + \theta(B)\varepsilon_t$$

ARIMA 모형

forecast 패키지 필요

```
library(forecast)
fit_arima <- auto.arima(ts_air)
summary(fit_arima)
checkresiduals(fit_arima)
forecast_arima <- forecast(fit_arima, h = 12)
plot(forecast_arima, main="Forecast using ARIMA")
```

6주차 과제

eclass 내 6주차 과제를 푸세요.

8-12번은 “Lec6_Nile.csv” 데이터를 활용하세요.